

Aryan Thakor — Modern Next.js Portfolio & CMS Studio

Complete Architecture, Full-Stack Migration, Dynamic Category Management & Delivery Audit

AUTHOR & DEVELOPER

Aryan Thakor

TECHNOLOGY STACK

Next.js 14 + Node.js

PRODUCTION STATUS

✓ 100% Live on Render

REPORT DATE

September 2026

48+

WEB PROJECTS

17+

GRAPHIC DESIGNS

6+

VIDEO PROJECTS

13/13

STATIC PAGES BUILT



Executive Summary

This document certifies the successful completion of the portfolio re-architecture for **Aryan Thakor** (Senior Web Developer, CMS Specialist & Graphic Designer). The project transformed a legacy React + Vite client single-page application into an ultra-fast, enterprise-grade **Next.js 14 Static Site Export** with an Express.js backend API and a stealth, two-factor authenticated **Admin CMS Studio**.

The system delivers exceptional lighthouse speed, zero-404 route handling, multi-channel category filtering, inline video playback, credential verification, and full category lifecycle management (Add, Edit, Delete, Restore) with seamless dual-layer synchronization across LocalStorage and backend JSON datastores.



Core Technology Stack

Layer	Technology	Version / Spec	Architectural Role
Framework	Next.js (Pages Router)	v14.2.35	Static HTML Pre-rendering, SSG exports, zero-server lag
Frontend UI	React & Vanilla CSS	v18.3.1	Curated dark/light design system with zero Tailwind runtime bloat
Icons & Assets	Lucide React	v0.428.0	Clean, scalable SVG vector iconography across all viewports
Backend API	Node.js & Express.js	v20.x / 4.x	REST endpoints for dynamic content, telemetry, and 2FA auth
Security	Passcode + TOTP 2FA	speakeasy / qrcode	RFC 6238 time-based authenticator with emergency recovery keys
Hosting	Render Cloud	Static Site + API	Global CDN deployment with automated CI/CD from GitHub main



Live Deployment URLs & Environment Metadata

Production Web Portal: <https://aryan-portfolio-s8il.onrender.com/>

GitHub Version Control: <https://github.com/Aryanthakor27/portfolio.git> (Commit 95dafce)

Secret Admin Access: <https://aryan-portfolio-s8il.onrender.com/admin/?key=aryan2026>

🌟 Major Portfolio Sections & Capabilities



Web Projects Showcase (48+ Sites)

Real-world commercial websites spanning luxury e-commerce, resorts, tech firms, industrial logistics, and healthcare. Includes real-time category filtering, external live site launchers, and tech stack tags.

E-CommerceHospitalityCorporateIndustrial



Graphic Design Showcase (17+ Works)

Curated visual branding, social media campaign posts, product photo manipulations, and commercial retouching. Features an instant high-resolution lightbox modal with zoom and details.

Logos & BrandingSocial PostsManipulation



Video Editing & Motion Graphics (6+ Projects)

High-retention video portfolio including commercial ads, short-form Reels/Shorts, YouTube documentary edits, and motion graphics. Features an embedded video playback modal with view metrics and tool badges.

CommercialsReels & ShortsMotion FX



Credentials & Letter of Experience

Verifiable career documentation including official Experience Letter from Digiva Inc / Rowwat Technologies CEO Rajni Patel & PM Jay Senjaliya, complete with embedded PDF viewer and direct download.

VerifiedPDF DownloadInternship Cert

📁 Advanced Category Management System

Delivered in the final sprint: A full-lifecycle **Category Manager** inside the Admin Dashboard that empowers the administrator to Add, Edit, Delete, and Restore categories without touching source code.

Operation	User Action	System Behavior & Data Integrity
 Add Category	Input label & auto-generated slug, click "+ Add Category"	Generates slug, verifies unique ID, saves to LocalStorage & dispatches cross-component update events.
 Edit Category	Click Edit (pencil) icon, modify label or slug, click "Update"	Updates category label. If slug changes, automatically cascades the new category ID to all existing assigned projects to prevent broken links!
 Delete Category	Click Delete (trash) icon	Calculates affected project count. Displays safety alert if projects are assigned. Removes category from filter tabs and cleans active filters.
 Restore Defaults	Click "Restore Defaults" button in category modal	Restores pristine default category arrays for Websites, Designs, or Videos upon confirmation.

🔒 Security & Admin CMS Architecture



Hidden Stealth Mode

Public navbar has zero visible admin lock or key icons. The administrative studio can only be reached via secret URL query param (`?key=aryan2026`) or direct route.



Multi-Factor Security (Passcode + 2FA)

Default Master Passcode protection (`2704`) paired with optional Google Authenticator TOTP 2FA and Emergency Recovery Keys to safeguard against unauthorized access.



Live Visitor Telemetry

Cloud-synced real-time visitor tracking via ntfy telemetry channel, capturing operating system, browser, timestamp, and active IP logs.



Recycle Bin Protection

Soft-delete architecture allowing deleted websites, graphic designs, and credentials to be restored with a single click before permanent purge.

⚡ Migration & Optimization Changelog

Item / Module	Previous State (Legacy)	Current State (Delivered)	Benefit
Build Architecture	React.js + Vite Client SPA	Next.js 14 Static HTML Export	Instant page loads, 0 client bundle delay, SEO meta prerendering
Obsolete Files	vite.config.js, index.html, App.jsx	Deleted & Cleaned	Clean repository footprint, no dead build dependencies
Navbar Lock Icon	Visible lock in desktop & mobile	Completely Removed	Stealth security, clean high-end public portfolio look
/designs Refresh Bug	Refreshing `/designs` redirected to home	Dedicated Route View Resolver	Direct browser reloads work on all routes with zero glitching
Video Stray Icon	Unwanted Film icon in header	Removed	Clean, balanced typographic section header
Graphic Design Tabs	Contained video-editing & reels	Purged & Separated	Graphic designs strictly contain branding, social, manipulation
Category Management	Create-only modal with no edit/delete	Complete CRUD Category Manager	Full autonomy to Add, Edit, Delete, and Restore all categories

🔧 Developer Handover & Operations Guide

1. Running the Project Locally:
`node backend/server.js` → Starts full-stack server on `http://localhost:5000/`
2. Building for Production:
`node build.js` → Installs dependencies, exports Next.js static bundle to `./dist` and `frontend/dist`
3. Deploying Updates to Render:
`git add -A && git commit -m "update message" && git push origin main` → Render triggers auto-build & deployment

✅ Final Certification & Sign-off

PROJECT STATUS: COMPLETED & SIGNED OFF

All user requirements, technical optimizations, Next.js conversions, stealth admin implementations, and category CRUD systems have been thoroughly tested, verified, and deployed to live production.

Aryan Thakor
Sr. Web Developer & Designer
Ahmedabad, India • 2026